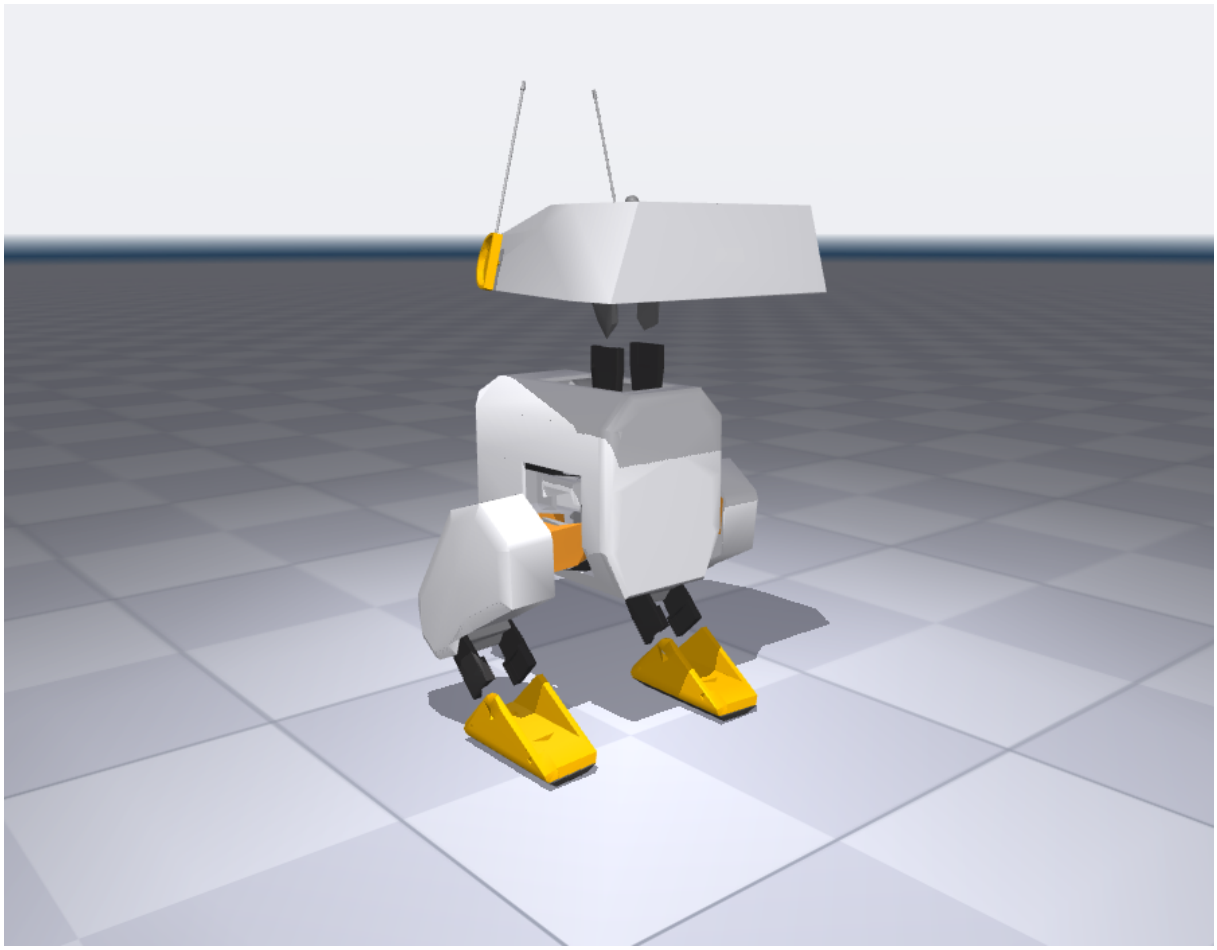
Micro Duck: Training Body

Open Duck Mini v2 simulation model (MuJoCo), checked and tweaked for training. Branch claude/micro-duck-sim, folder duckbody/.



Standing in the home pose, 3/4 view

Specs

	Simulation model	Real build plan
Weight	2.107 kg	about 2 kg (within 2.5%)
Height	395 mm standing, 430 mm legs straight, 483 mm to antenna tips	about 42 cm
Motors	14: per leg hip yaw, hip roll, hip pitch, knee, ankle; neck pitch, head pitch, yaw, roll	14 x STS3215 + 2 antenna servos
Leg	thigh 78.7 mm, shin 78.7 mm, ankle to sole 38.8 mm	
Foot	103 x 41 mm TPU sole, 8 mm	same part
Stance	hips 70 mm apart; feet cover 190 x 92 mm	
Standing pose	base 161 mm, centre of mass 205 mm, knees 78 deg	
Weight split	trunk 0.70 kg, head 0.41 kg (19%), each leg about 0.41 kg	

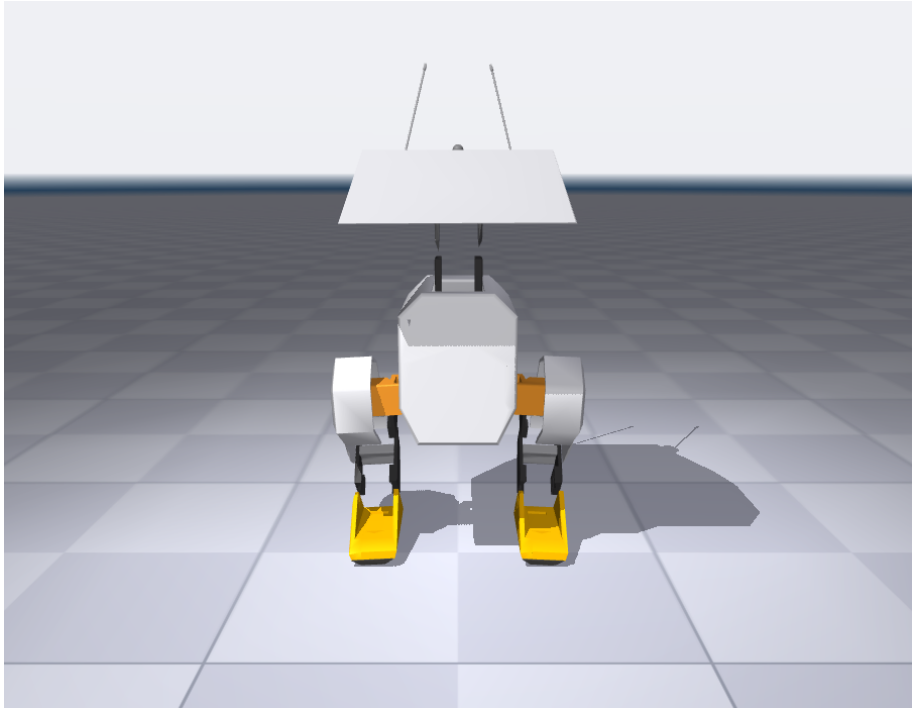
Standing effort	knees 0.44-0.50 N m, others under 0.2 N m (26% of servo stall)	

Joint ranges

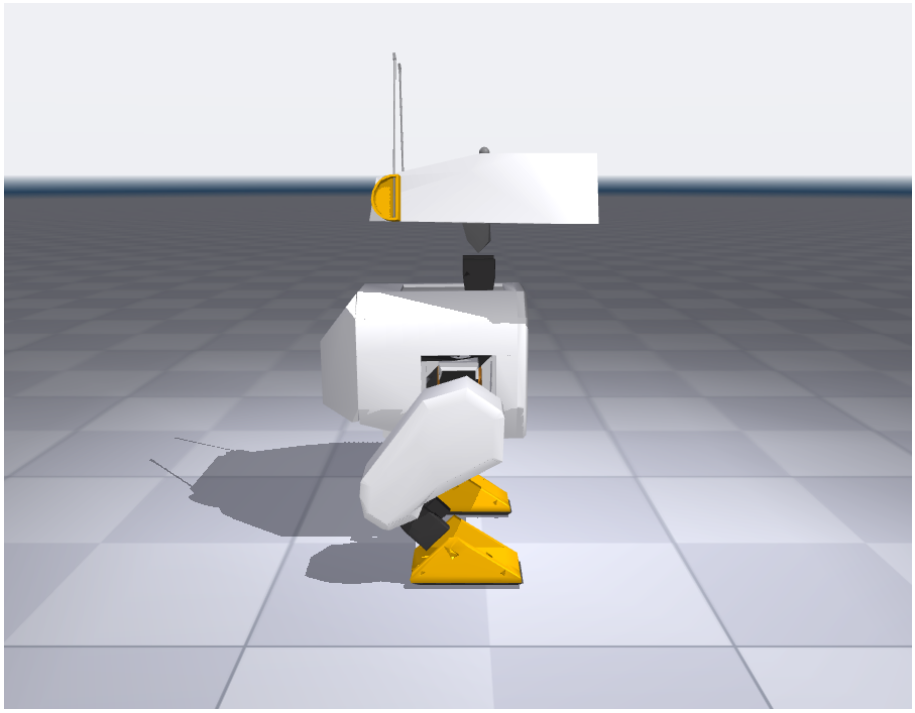
Joint	Range (deg)
Hip yaw	-30 to 30
Hip roll	-25 to 25
Hip pitch	-70 to 30 (mirrored on right)
Knee	-90 to 90 (proposed: up to 125)
Ankle	-90 to 90
Neck pitch	-20 to 65
Head pitch	-45 to 45
Head yaw	-160 to 160
Head roll	-30 to 30

Stand test

10 runs of 5 s from slightly wrong joint angles and up to 3 deg tilt: 0 falls, worst tilt 4.4 deg, settles under 0.3 deg, height steady within about 1 mm.



Front view



Side view

Fixed automatically

Problem	Fix
Only the soles could touch anything; body and head passed through the floor	22 real printed shells now collide (no added weight)
Hard stops at the servo limit carried the robot for free	Hard stop moved 5 deg further out; the servo must hold the pose
Each start began 17 mm inside the floor	Start height computed so the feet just touch
Balance sensor 12 mm from the real chip	Moved to the BNO055 position

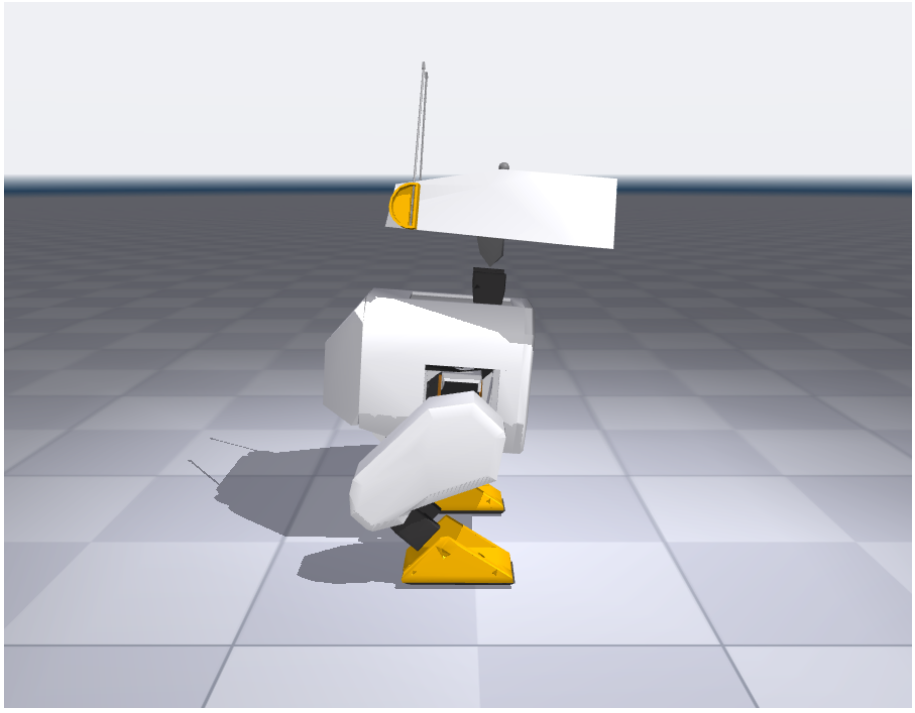
For you to decide

Tweak	Why	Suggestion
Knee range 125 deg	90 deg allows only a 12 mm crouch; sleep and sit poses are impossible	Measure a real leg first; use contact angle minus 10 deg
Servo limit 1.91 N m	Sim allows 3.23 N m; the 7.4 V STS3215 gives 1.91	Check servo label; training varies 1.9-3.23 anyway
Real part weights	CAD parts are solid; prints at 15% infill are lighter	Weigh parts while building
Gear backlash	Cheap gears have about 0.5 deg play	Switch on for the last walking stage

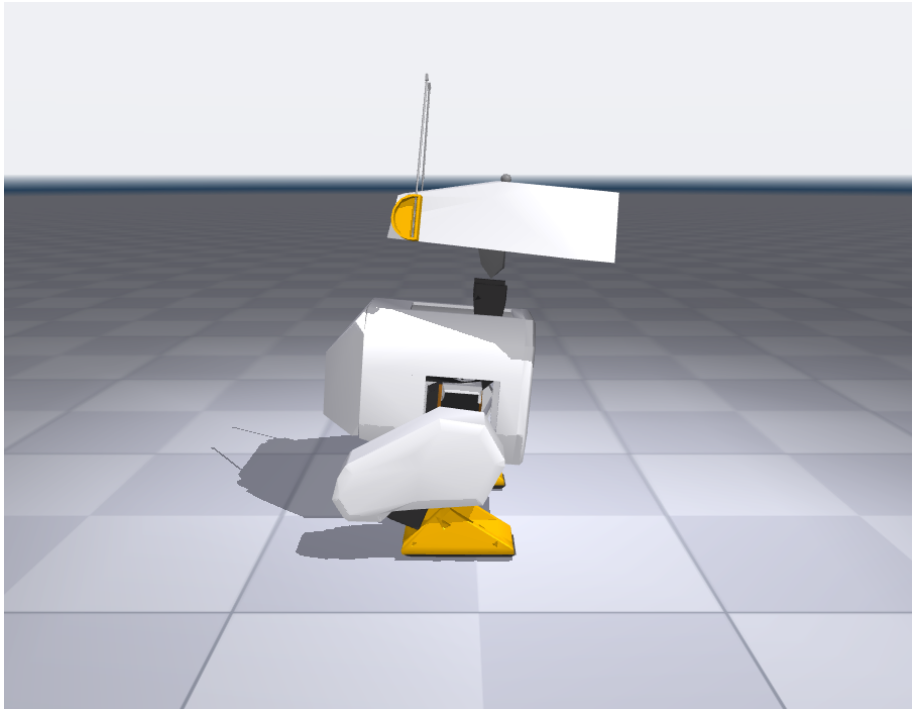
Design notes

Top-heavy head (19% of weight, high up) hits first in every fall: add TPU bumpers on head and back corners, no extra electronics in the head. Weight sits toward the heels (31 mm to heel edge, 61 mm to toe): backward pushes tip it sooner, so training uses more of them.

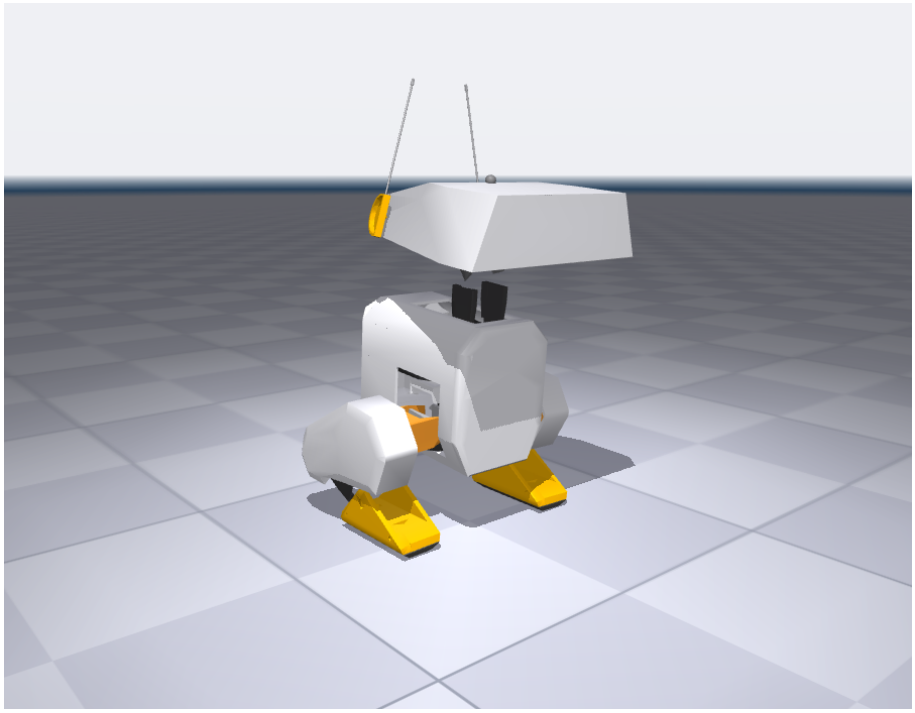
Crouch and collision views



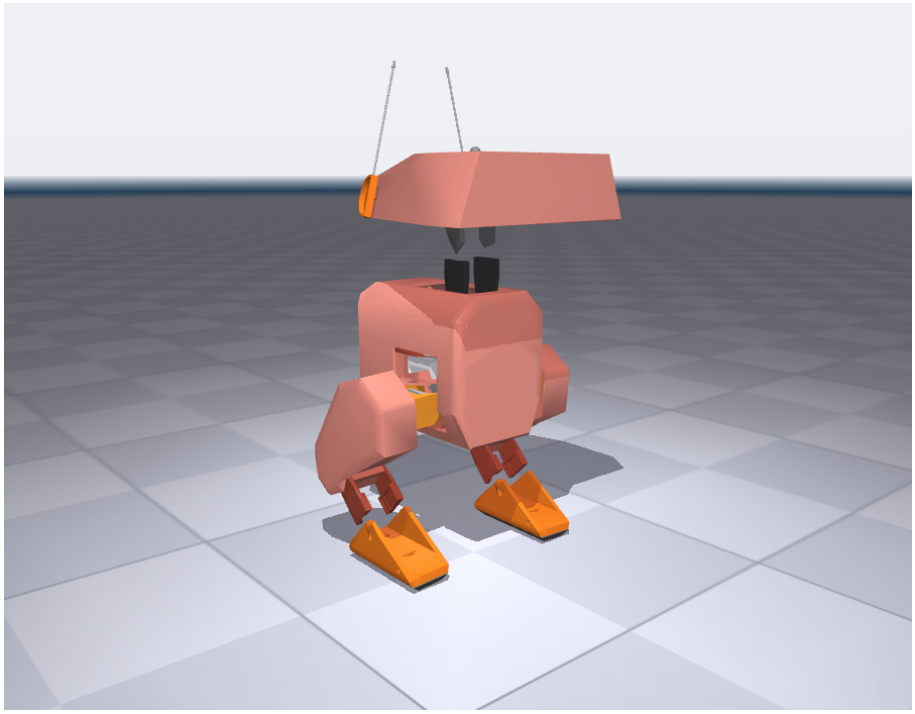
Deepest crouch with the current 90 deg knee limit



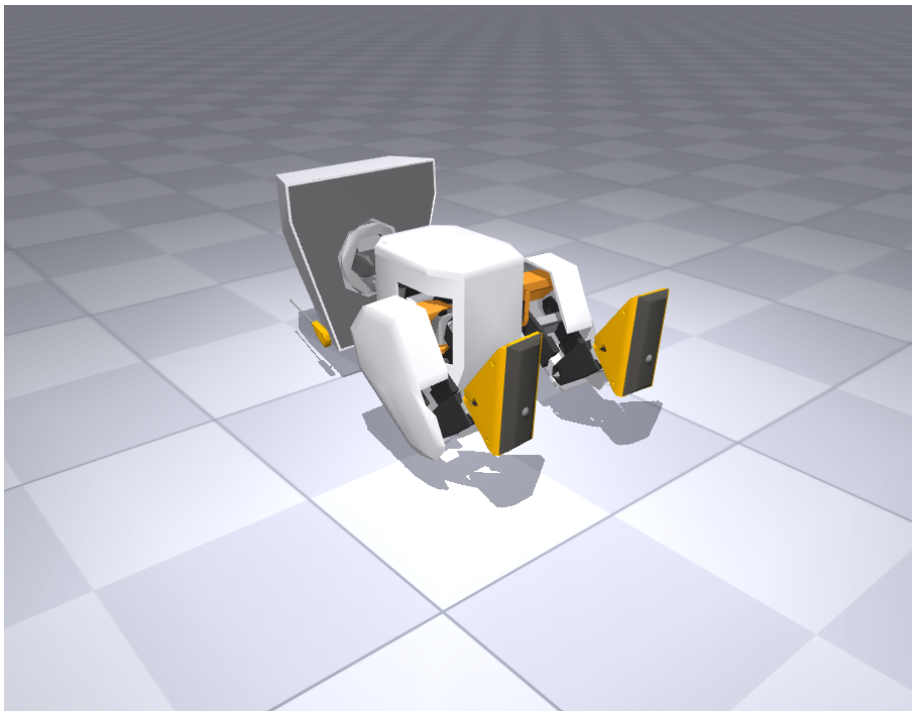
Proposed 125 deg knee range: body sits 46 mm lower



Proposed 125 deg crouch, 3/4 view



Collision shells added so the body can touch the ground



Lying on its back, resting on the new shells